

9(a)	<i>eight from:</i> 1 double membrane ; <i>inner membrane</i> 2 folded / cristae ; 3 increased / large, surface area ; 4 has, ATP synthase / stalked particles ; 5 has, carrier (proteins) / cytochromes ; 6 (site of) ETC / oxidative phosphorylation / chemiosmosis ; <i>intermembrane space</i> 7 has low pH / high concentration of protons ; 8 accepts protons from ETC / AW ; 9 proton gradient between intermembrane space and matrix or - protons move from intermembrane space to matrix ; 10 <i>ref. to</i> ATP synthesis ; <i>matrix</i> 11 contains enzymes ; 12 site of, link reaction / the Krebs cycle ; <i>outer membrane</i> 13 presence of carriers for, pyruvate / reduced NAD ; 14 AVP ; e.g. ribosomes / DNA, plus function	8
------	---	---

Question	Answer	Marks
9(b)	<i>seven from:</i> <i>liver cells (max 6)</i> 1 only glycolysis occurs ; 2 pyruvate, cannot enter mitochondrion / remains in the cytoplasm ; 3 (pyruvate) becomes, hydrogen acceptor / reduced ; 4 by reduced NAD (from glycolysis) ; 5 lactate produced ; 6 lactate dehydrogenase ; 7 production of, 4 ATP / 2 ATP / small amount of ATP ; 8 allows glycolysis to continue ; <i>yeast cells</i> 9 decarboxylation / CO₂ removed ; 10 ethanal becomes, hydrogen acceptor / reduced ; 11 two steps (instead of one) ; 12 irreversible reaction (instead of reversible) ; 13 ethanol dehydrogenase ; 14 ethanol produced ;	7